

Target Journal Declaration

Target: *IEEE Journal of Biomedical and Health Informatics* (JBHI) **Second choice:** *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (TNSRE) **Manuscript format:** IEEE two-column journal format (`\documentclass[journal]{IEEEtran}`), which is the format JBHI requires for submission.

Why JBHI fits this paper

Scope match. JBHI publishes informatics and machine-learning methods applied to biomedical and health data, with an explicit emphasis on work that connects computational method to clinical use. This manuscript is exactly that: a multimodal model that reads the full clinical polysomnogram and produces two clinically actionable outputs — the sleep stage of every 30-second epoch and a per-epoch respiratory-event label — evaluated on a patient cohort rather than a healthy convenience sample.

Clinical population, not a benchmark dataset. The cohort is iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients). The paper’s central finding is clinical rather than architectural: deep architectures developed on healthy sleep lose 15–20 accuracy points on this population, while a compact, physiologically-grounded model does not. JBHI’s readership — clinical informatics researchers and biomedical engineers — is the audience for whom that finding changes practice. A general machine-learning venue would treat it as a negative result; JBHI treats it as a deployment constraint.

Multi-signal physiological data is core JBHI territory. The model consumes EEG, EOG, EMG, ECG, airflow, thoracic and abdominal effort, and SpO2. JBHI regularly publishes multimodal physiological-signal work, including sleep staging and sleep-disordered-breathing detection, so both the signal modality and the task have established precedent in the journal.

Clinical validation is expected, not optional. The manuscript reports association with clinically scored AHI (Spearman $\rho = 0.315$, $p = 0.0017$, $n = 96$), staging accuracy stratified by AASM severity band, and per-event-type detection (hypopnea 0.692, obstructive 0.763, central 0.840). These clinical-facing analyses are the kind JBHI reviewers ask for and many methods venues consider out of scope.

Interpretability framed as physiological attribution. The modality-ablation grid demonstrates a clean dissociation — cardiorespiratory channels drive the respiratory head (AUC 0.711 $\rightarrow$ 0.673 on removal, $p = 0.004$) while contributing nothing to staging ($p = 0.91$). JBHI’s clinical readership requires that a model’s behaviour be explainable in physiological terms rather than by saliency heatmaps alone.

Why not the alternatives

TNSRE (second choice) is a strong fit for the neural-signal processing, but its centre of gravity is neural engineering, rehabilitation and neural interfaces. The respiratory and cardiovascular half of this paper sits outside that emphasis, and the joint staging-plus-breathing formulation is the contribution we most want reviewed.

****Elsevier’s *Computers in Biology and Medicine***** is a reasonable scope match and was considered. We prefer JBHI because its reviewer pool is more specialised in physiological time-series and sleep medicine, and because the stroke-cohort framing is more likely to be evaluated on clinical merit there.

General ML venues (NeurIPS/ICLR-style) are not appropriate. The architecture is a considered recombination of established components rather than a novel learning algorithm, and the paper’s value is the clinical finding and the cohort — which such venues systematically undervalue.

Format compliance

The manuscript is prepared in IEEEtran two-column journal style, matching JBHI's author requirements: structured abstract, IEEE keyword block, IEEE numeric citation style, two-column floats, and the corresponding-author and declaration blocks (data availability, code availability, ethics, funding) required at submission.